Density operator: An equivalent formulation of QM in terms of density matrices rather than state vectors.

Suppose a quantum system in multiple states $|\psi_i\rangle$ with probability p_i

$\{p_i, |\psi_i\rangle\}$ define an ensemble of pure states. (probabilistic mixture of states) which are orthonormal to each other.

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$$

Unitary evolution in density matrix language

$$|\psi_i\rangle \xrightarrow{U} U|\psi_i\rangle$$

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i| \to \sum_i p_i U|\psi_i\rangle\langle\psi_i|U^\dagger = U\left(\sum_i p_i|\psi_i\rangle\langle\psi_i|\right)U^\dagger$$

$$\rho \Rightarrow U\rho U^\dagger$$

Measurements:

if the initial state was $|\psi_i\rangle$

$$p(m|i) = \langle\psi_i|M_m^\dagger M_m|\psi\rangle = tr(M_m^\dagger M_m|\psi_i\rangle\langle\psi_i|)$$

$$p(m) = \sum_i p(m|i)p_i = \sum_i p_i\, tr(M_m^\dagger M_m|\psi_i\rangle\langle\psi_i|) = tr(M_m^\dagger M_m \underbrace{\sum_i p_i|\psi_i\rangle\langle\psi_i|})$$

$$= tr(M_m^\dagger M_m\rho)$$

If the system is characterized by a density matrix then $p(m) = tr(M_m^\dagger M_m \rho)$

In state vector formulation $p(m) = \langle\psi|M_m^\dagger M_m|\psi\rangle$

What is the density matrix after measurement.

Given $|\psi_i\rangle$ was the initial state.

$$|\psi_i^m\rangle = \frac{M_m|\psi_i\rangle}{\sqrt{\langle\psi_i|M_m^\dagger M_m|\psi\rangle}}$$

After measurement we have an ensemble of states $|\psi_i^m\rangle$ $\{p(i|m), |\psi_i^m\rangle\}$.

$$\rho_m = \sum_i p(i|m)|\psi_i^m\rangle\langle\psi_i^m| = \sum_i p(i|m)\frac{M_m|\psi_i\rangle\langle\psi_i|M_m^\dagger}{\boxed{\langle\psi_i|M_m^\dagger M_m|\psi_i\rangle}}$$

$$\frac{p(i|m)}{p(m)} = \frac{p(m|i)p_i}{p(m)}$$

$$\rho_m = \sum_i \frac{M_m|\psi_i\rangle\langle\psi_i|M_m^\dagger}{tr(M_m^\dagger M_m\rho)}p_i \qquad \sum_i p_i|\psi_i\rangle\langle\psi_i| = \rho$$

$$\rho_m = \frac{M_m\rho M_m^\dagger}{tr(M_m^\dagger M_m\rho)}$$

∴ Basic postulates of QM can be re-written in the language of density matrices.

Can we write the density matrix without the state vector.

A quantum system in state $|\psi\rangle$, which is known

$|\psi\rangle$ is a pure state.

$\rho = |\psi\rangle\langle\psi|$

Otherwise it is a mixed state. A probabilistic mixture of pure states.

A pure state $|\psi\rangle = a|0\rangle + b|1\rangle = \underbrace{\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)}_{A}$

$\rho = |\psi\rangle\langle\psi|$

$$\rho = \underbrace{\left(\tfrac{1}{2}\right)|0\rangle\langle0| + \left(\tfrac{1}{2}\right)|1\rangle\langle1|}_{B}$$

if I measure system A $p(|0\rangle) = \frac{1}{2}$, $p(|1\rangle) = \frac{1}{2}$

for system B:

$p(|0\rangle) = \frac{1}{2}$

$p(|1\rangle) = \frac{1}{2}$

system A is a superposition of pure states (basis states) system A the state $|0\rangle$ and $|1\rangle$ simultaneously. However system B is a probabilistic mixture.

⇒ for system A $\rho = |\psi\rangle\langle\psi|$

$$\rho = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)\left(\frac{1}{\sqrt{2}}(\langle0| + \langle1|)\right)$$

$$\rho = \frac{1}{2}(|0\rangle\langle0| + |0\rangle\langle1| + |1\rangle\langle0| + |1\rangle\langle1|)$$

Properties of density matrix.

$\sum_i p_i = 1$ $\qquad \rho = \sum_i \underbrace{p_i}|\psi_i\rangle\langle\psi_i|$

$tr(\rho) = 1.$ $\qquad \rho = \sum_j \lambda_j |j\rangle\langle j|$

A pure state satisfies $tr(\rho^2) = tr(\rho) = 1.$

$\rho = |\psi\rangle\langle\psi|$ $\qquad \langle\psi|\psi\rangle = 1$

$\rho^2 = |\psi\rangle\langle\psi|$

$tr(\rho^2) = tr(\rho) = 1.$

$tr(\rho^2) = 1 \Rightarrow$ the system is in a pure state.

$tr(\rho^2) < 1 \Rightarrow$ system is in a mixed state.

$$\rho = \sum_i p_i|\psi_i\rangle\langle\psi_i|$$

$$\rho^2 = \sum_{ij} p_i p_j |\psi_i\rangle\underbrace{\langle\psi_i|\psi_j\rangle}_{\delta_{ij}}\langle\psi_j|$$

$$= \sum_i p_i^2|\psi_i\rangle\langle\psi_i|$$

$\sum_i p_i = 1$ from $tr(\rho) = 1.$

$p_i^2 < p_i$

$tr(\rho^2) = \sum_i p_i^2 < 1$

$\Rightarrow tr(\rho^2) < 1 \Rightarrow$ system is in a mixed state.

Ensemble of pure states. $|\psi_i\rangle$

$\rho_i = |\psi_i\rangle\langle\psi_i|$

$\rho = \sum p_i\rho_i$ (probabilistic mixture of density matrices)

$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$

1. $tr(\rho) = 1$
2. ρ is positive definite

$|\varphi\rangle$

$$\langle\varphi|\rho|\varphi\rangle = \sum_i p_i\underbrace{|\langle\varphi|\psi_i\rangle|^2}$$

$$= \sum_i p_i \text{ (positive real number)}$$

$$\geq 0$$

3. $\rho = \sum_i p_i|\psi_i\rangle\langle\psi_i|$

$(|\psi_i\rangle\langle\psi_i|)^\dagger = |\psi_i\rangle\langle\psi_i|$

ρ is hermitian.

$tr(\rho) = 1$ $\qquad tr(|a\rangle\langle b|) = \langle b|a\rangle$

$\rho = \sum_i p_i|\psi_i\rangle\langle\psi_i|$

$$tr(\rho) = \sum_i p_i\, tr(|\psi_i\rangle\langle\psi_i|) = \sum_i p_i \underbrace{\langle\psi_i|\psi_i\rangle}_{=1} = \sum_i p_i = 1$$

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1. ρ_i with p_i

$\rho = \sum_i p_i\rho_i$

2. $\rho' \to U\rho U^\dagger$ where U is unitary.

$\rho'(t_2) = U(t_1,t_2)\rho(t_1)U^\dagger(t_1,t_2)$

3. $p(m) = tr(M_m^\dagger M_m\rho)$

post-measurement ρ

$$\frac{M_m\rho M_m^\dagger}{tr(M_m^\dagger M_m\rho)}$$

4. Composite system

$\rho = \rho_1\otimes\rho_2\otimes \ldots \otimes\rho_n$

Suppose a single qubit system

$\rho = \frac{3}{4}|0\rangle\langle0| + \frac{1}{4}|1\rangle\langle1|$ ✓

$|a\rangle = \sqrt{\frac{3}{4}}|0\rangle + \sqrt{\frac{1}{4}}|1\rangle$

$|b\rangle = \sqrt{\frac{3}{4}}|0\rangle - \sqrt{\frac{1}{4}}|1\rangle$

$\rho = \frac{1}{2}|a\rangle\langle a| + \frac{1}{2}|b\rangle\langle b|$

$\rho = \frac{3}{4}|0\rangle\langle0| + \frac{1}{4}|1\rangle\langle1|$

System described by $|a\rangle, |b\rangle$ is very different from system $|0\rangle, |1\rangle$

$\{\frac{1}{2}, \frac{1}{2}, |a\rangle, |b\rangle\}$ Ensemble A

$\{\frac{3}{4}, \frac{1}{4}\ |0\rangle, |1\rangle\}$ Ensemble B

Ensemble A is unitarily equivalent to ensemble B.

Given two ensembles $|\psi_i\rangle, |\varphi_i\rangle$ these ensembles will generate the same density matrix if and only if

$$|\psi_i\rangle = U_{ij}|\varphi_j\rangle$$

U_{ij} is unitary

$$\rho = \sum_i p_i|\psi_i\rangle\langle\psi_i| = \sum_j q_j|\varphi_j\rangle\langle\varphi_j|$$

this implies

$$\sqrt{p_i}|\psi_i\rangle = \sum_j U_{ij}\sqrt{q_j}|\varphi_j\rangle$$

Reduced density matrices tool for subsystem.

A and B

$A\otimes B$

ρ^{AB}

$\downarrow$

partial trace

$\rho_A = tr_B(\rho^{AB}) \Rightarrow$ partial trace over system B

$$tr_B(|a_1\rangle\langle a_2|\otimes \underbrace{|b_1\rangle\langle b_2|})$$

$$= |a_1\rangle\langle a_2|\, tr(|b_1\rangle\langle b_2|)$$

$$= |a_1\rangle\langle a_2|(\langle b_2|b_1\rangle)$$

$= |a_1\rangle\langle a_2|$ over system A

$|a_1\rangle$ $|a_2\rangle$ are any two vectors in the state space of A

$|b_1\rangle$ $|b_2\rangle$

Consider $\rho^{AB} = \rho\otimes\sigma$

$\rho^A = tr_B(\rho\otimes\sigma) = \rho\, tr(\sigma) = \rho$

$\rho^B = tr_A(\rho\otimes\sigma) = tr(\rho)\sigma = \sigma$

Consider the Bell state $|\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$ ✓

$\rho = |\psi\rangle\langle\psi|$ ✓

$$= \frac{(|00\rangle + |11\rangle)}{\sqrt{2}}\frac{(\langle00| + \langle11|)}{\sqrt{2}}$$

$$\rho = \frac{|00\rangle\langle00| + |00\rangle\langle11| + |11\rangle\langle00| + |11\rangle\langle11|}{2}$$

A two qubit system $|00\rangle = |0\rangle^1\otimes|0\rangle^2$

Tracing out the second qubit

$$\rho' = tr_2(\rho) = \frac{|0\rangle\langle0| + |1\rangle\langle1|}{2}$$

$$tr_2(|00\rangle\langle00|) = tr_2(|0^1\rangle\otimes|0^2\rangle\langle0^1|\otimes\langle0^2|) = |0^1\rangle\langle0^1|\, tr_2(|0^2\rangle\langle0^2|) \quad \langle0^2|0^2\rangle$$

$$= |0\rangle\langle0|$$

$$tr_2(|00\rangle\langle11|) = tr_2(|0^1\rangle\otimes|0^2\rangle\langle1^1|\otimes\langle1^2|) = |0^1\rangle\langle1^1|\, tr_2(|0^2\rangle\langle1^2|) \quad \langle0|1\rangle$$

$$= 0$$

$$\rho_1 = \frac{|0\rangle\langle0| + |1\rangle\langle1|}{2} = \frac{1}{2}\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix} = \frac{I}{2}$$

ρ was pure state. (joint system was a pure state) but the first qubit is in a mixed state.

$\rho_1 = \frac{I}{2}$

$\rho_1^2 = \frac{I}{4}$

$tr(\rho_1^2) < 1$